CADET Expert System

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Introduction to CADET

- CADET: Cancer Detection Expert System
  - Rule-based medical decision support system
  - Designed for early cancer detection
  - Assists doctors, not replaces them

CADET Architecture Overview

- Inference Engine
- Knowledge Base
- Working Memory
- User Interface
- Explanation Module

Knowledge Base (Detailed)

- Hundreds of IF–THEN medical rules
  - Derived from oncologists & clinical guidelines
  - Symptoms, biopsy results, imaging findings
  - Risk factors (age, genetics, lifestyle)

Working Memory

- Stores patient-specific facts
- Updated dynamically during consultation
- Used by inference engine for reasoning

Inference Engine

- Core reasoning component
  - Mostly Forward Chaining
  - Can support Backward Chaining
  - Matches facts with rules

Forward Chaining in CADET

- Facts → Rules → Conclusion
  - Data-driven reasoning
  - Starts with patient data
  - Fires matching rules automatically

Backward Chaining (Optional Mode)

- Goal → Required Facts → Verification
  - Used for hypothesis testing
  - Efficient when checking specific cancer type

Case Study: Patient Input

- Age: 52
  - Family History: Yes
  - Breast Lump: Present
  - Biopsy: Suspicious
  - Lymph Nodes: Enlarged

Example Rule 1

- IF age > 50
  - AND breast lump present
  - THEN Moderate Cancer Risk

Example Rule 2

- IF biopsy positive
  - AND lymph nodes enlarged
  - THEN High Cancer Probability

Rule Firing Process

- Patient facts stored
  - Rule 1 conditions satisfied → Moderate risk
  - Rule 2 conditions satisfied → High probability
  - Final Output: High Risk of Breast Cancer

System Output

Diagnosis: High Risk

Confidence Level: High

Recommendation: Oncology referral

Suggested Tests: MRI, Advanced biopsy

Explanation Facility

- Explains why diagnosis was made
  - Shows which rules fired
  - Increases doctor trust
  - Enhances transparency

Flow of Reasoning

Patient Data Entered

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Stored in Working Memory

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Rules Matched & Fired

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Risk Level Generated

Advantages

- Early detection support
  - Structured reasoning
  - Reduces diagnostic variability
  - Handles large medical knowledge

Technical Advantages

- Modular architecture
  - Easy rule addition
  - Transparent reasoning process
  - Deterministic decision logic

Limitations

- Needs frequent updating
- Dependent on rule quality
- Cannot learn automatically
- Not suitable for rare unknown cases

Real-World Deployment Challenges

- Medical validation required
  - Doctor training needed
  - Integration with hospital systems
  - Continuous knowledge updating